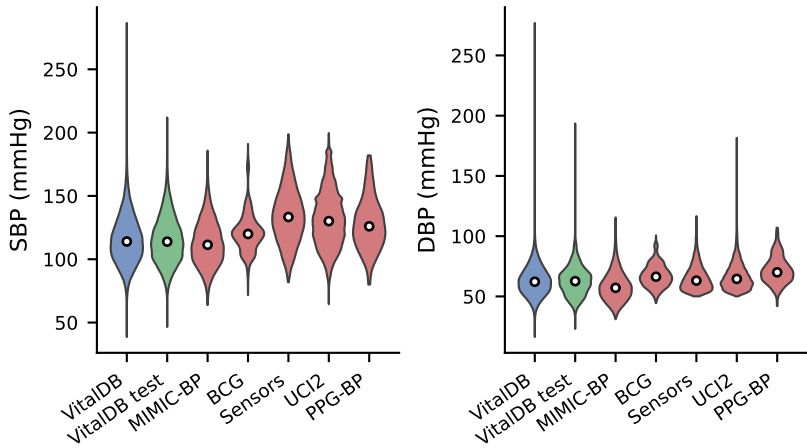


**a**

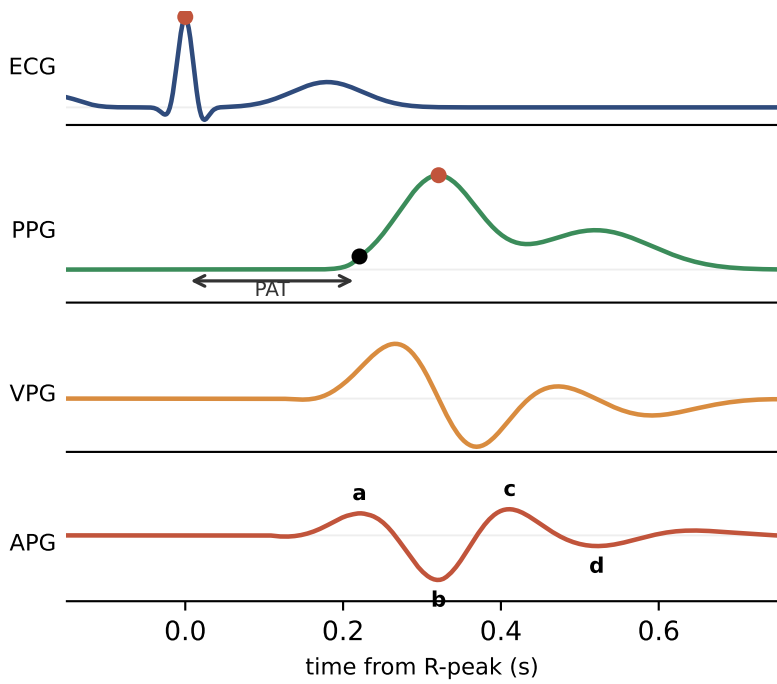


**b**

Datasets

| Dataset      | Role  | Seg  | Subj   | SBP/DBP | Age [5-95] | Chan    |
|--------------|-------|------|--------|---------|------------|---------|
| VitalDB      | train | 465k | 1,293  | 115/63  | 61 [31-79] | ECG+PPG |
| VitalDB test | ID    | 58k  | 144    | 115/63  | 60 [30-79] | ECG+PPG |
| MIMIC-BP     | OOD   | 137k | 1,524  | 113/58  | --         | ECG+PPG |
| BCG          | OOD   | 3k   | 40     | 121/67  | 28 [19-65] | PPG     |
| Sensors      | OOD   | 11k  | 1,195  | 134/65  | 57 [33-81] | PPG     |
| UCI2         | OOD   | 411k | 10,793 | 132/67  | --         | PPG     |
| PPG-BP       | OOD   | 619  | 218    | 128/72  | 58 [25-80] | PPG     |

**c** Feature extraction (ECG, PPG, 1st/2nd deriv.)

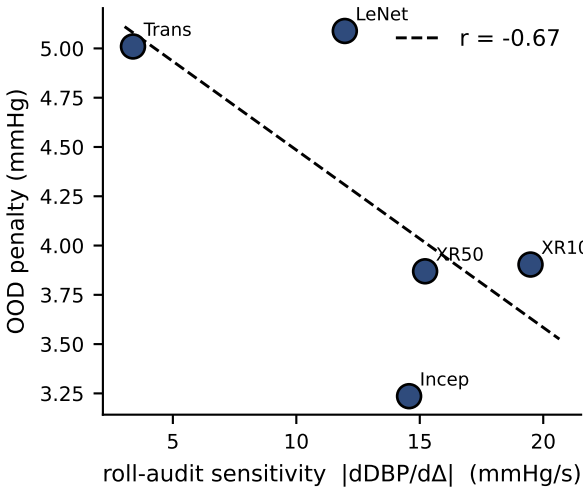


**d**

DBP MAE (mmHg): ID VitalDB, OOD MIMIC-BP

| Model                      | size    | ID   | OOD   |
|----------------------------|---------|------|-------|
| LeNet1d                    | 659k    | 8.5  | 13.5  |
| Inception1d                | 472k    | 8.6  | 11.8  |
| XResNet50                  | 887k    | 8.7  | 12.6  |
| XResNet101                 | 1.8M    | 8.6  | 12.5  |
| Transformer                | 107k    | 8.6  | 13.6  |
| LightGBM                   | ~8k     | 8.2  | 15.1  |
| LightGBM (83 feat)         | 50k lv  | 8.10 | 14.65 |
| LightGBM + demographics    | 50k lv  | 8.28 | 16.76 |
| LightGBM single tree       | 64 lv   | 8.92 | 12.24 |
| LightGBM PAT only          | 6 feat  | 8.47 | 11.09 |
| Faithful: PAT + morphology | 4.5k lv | 8.63 | 15.74 |
| LightGBM 12 feat, 20 trees | 620 lv  | 8.69 | 15.73 |

**e** Mechanism vs OOD robustness



**f**

Decodability is flat across models (number = |roll slope|, mmHg/s)

